

Name: _____

Date: _____

LEARNING TARGET

I can investigate and record how actions physically change everyday materials.

KEY STRATEGY + WORKED EXAMPLE

Predict, test one action safely, observe closely, record what changed, and decide whether the material stayed the same kind of material.

Fair-test idea: Test one action at a time. Bend foil, stretch a rubber band gently, twist paper, squash playdough, and mix sand with water. Do not taste materials.

A. TRY TOGETHER - USE THE STRATEGY

1. Which action curves a material: bend or mix?
2. Which action pulls a material longer: stretch or squash?
3. Predict what happens when foil is bent.
4. Name one safety rule for stretching a rubber band.

B. CORE PRACTICE - SHOW YOUR THINKING

5. Paper: can it bend?

6. Rubber band: can it stretch?

7. Playdough: can it squash?

8. Foil: can it twist?

9. After bending foil, is it still foil?

10. After tearing paper, is it still paper?

11. Sand + water: can you still see both materials?

12. Which observation should be measured or described?

C. INDEPENDENT PRACTICE - COMPLETE AND CHECK

13. Sort action: bend / stretch / twist / squash - rolling playdough flat

14. Sort action - turning a pipe cleaner around a pencil

15. Sort action - pulling elastic longer

16. Sort action - curving cardboard

17. Change that can often reverse: stretched elastic or torn paper?

18. Change that is hard to reverse: bent sponge or torn paper?

19. Complete: A physical change can alter the ____ of a material.

20. Write one observation using 'I noticed...'

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D. APPLY THE SKILL

21. Plan a safe test comparing how paper and foil respond to bending. Include prediction, same action and observation table.

22. Mix one spoon of sand with three spoons of water. Record what you see immediately and after settling. Do not taste.

23. Choose the best material for a bendable bookmark: foil, glass or dry biscuit. Justify using properties and test evidence.

E. REASON, CHECK AND CORRECT

24. Why must we use the same action when comparing two materials?

25. Correct: 'If a material changes shape, it becomes a completely new material.'

EXIT TICKET + CHALLENGE

26. State one material, one action, the observed change and whether it could be reversed.

Challenge: Design a safe investigation testing two actions on two materials. Include a fair-test rule and recording table.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.

27. Explain one strategy you used today.

28. Check one answer in a different way.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: _____

31. The most difficult part was: _____

Why was it difficult? _____

32. My next practice step is: _____

TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

A. 1) bend. 2) stretch. 3) It curves/folds/changes shape. 4) Keep away from face; stretch gently; inspect for damage.

B. 5) yes. 6) yes. 7) yes. 8) yes. 9) yes. 10) yes. 11) yes. 12) Shape, length, texture, appearance or ability to return.

C. 13) squash. 14) twist. 15) stretch. 16) bend. 17) stretched elastic. 18) torn paper. 19) shape/form. 20) Observation varies and must describe evidence.

D. 21) Same-sized samples and same bend; record amount/shape and return. 22) Sand disperses then settles; both materials remain observable. 23) Foil, because it bends without breaking; evidence required.

E. 24) It makes the comparison fair. 25) False; many physical changes alter shape without changing material composition.

EXIT. 26) Answers vary; all four elements required.

Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.

F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may think every shape change creates a new material or every material returns to its original shape. Compare reversible and irreversible physical changes.

TEACHER CHECK

[] Uses an appropriate Year 2 Science strategy.

[] Shows or explains thinking clearly.

[] Checks a response using evidence, a rule, or a second method.

[] Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.